

UNIVERSIDAD CENTRAL DEL ECUADOR
FACULTAD FILOSOFÍA, LETRAS Y CIENCIAS DE LA EDUCACIÓN
CARRERA PEDAGOGÍA DE LAS CIENCIAS EXPERIMENTALES
INFORMÁTICA

Nombre: Pujota Sandy

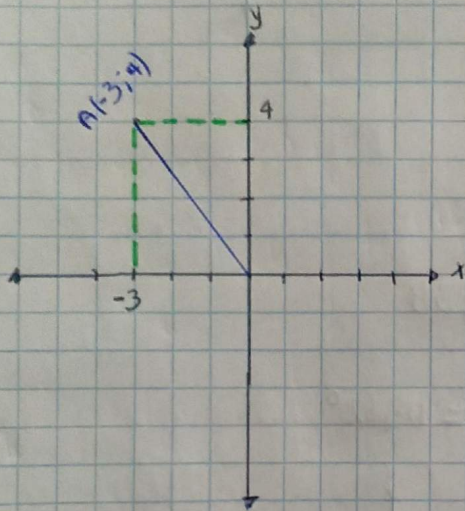
Curso: 5^{to} "B"

Fecha: 29/04/2026

Asignatura: Física Aplicada

Indicaciones: Realizar lo ejercicios en 2D.

1- $\vec{A} = -3\vec{i} + 4\vec{j}$ [m]



$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(-3)^2 + (4)^2}$$

$$|\vec{A}| = \sqrt{9 + 16}$$

$$|\vec{A}| = \sqrt{25} = 5 \text{ [m]}$$

Valor Unitario

$$\vec{U}_A = \frac{A_x \vec{i} + A_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{-3\vec{i}}{5} + \frac{4\vec{j}}{5}$$

$$\vec{U}_A = 0,6\vec{i} + 0,8\vec{j}$$

$$|\vec{U}_A| = \sqrt{(0,6)^2 + (0,8)^2}$$

$$|\vec{U}_A| = 1$$

Buscar θ Dirección

$$\theta = ?$$

$$\tan \theta = \frac{A_y}{A_x}$$

$$\theta = \tan^{-1} \left(\frac{A_y}{A_x} \right)$$

$$\tan^{-1} \left(\frac{4}{-3} \right) = -53,13^\circ$$

$$\theta = 180 - 53,13 = 126,87^\circ$$

$$2. \vec{A} = (-5\vec{i} + 12\vec{j})$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(-5)^2 + (12)^2}$$

$$|\vec{A}| = \sqrt{25 + 144}$$

$$|\vec{A}| = \sqrt{169} = 13[m]$$

Valor Unitario

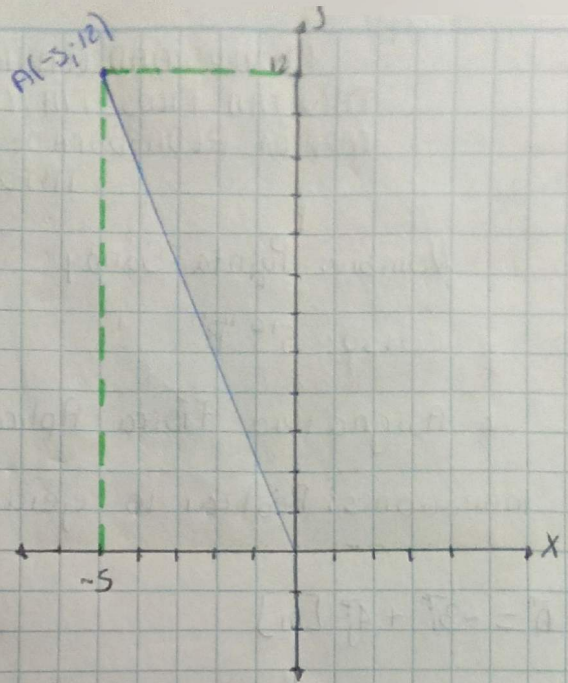
$$\vec{U}_A = \frac{A_x \vec{i} + A_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{-5\vec{i}}{13} + \frac{12\vec{j}}{13}$$

$$\vec{U}_A = -0,38\vec{i} + 0,92\vec{j}$$

$$|\vec{U}_A| = \sqrt{(-0,38)^2 + (0,92)^2}$$

$$|\vec{U}_A| = 1$$



Dirección θ ?

$$\tan \theta = \frac{A_y}{A_x}$$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

$$\tan^{-1}\left(\frac{12}{-5}\right) = -67,38$$

$$\theta = 180 - 67,38 = 112,62^\circ$$

$$3. \vec{A} = (9\vec{i} + 12\vec{j})$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(9)^2 + (12)^2}$$

$$|\vec{A}| = \sqrt{81 + 144} = \sqrt{225}$$

$$|\vec{A}| = 15 [m]$$

Valor Unitario

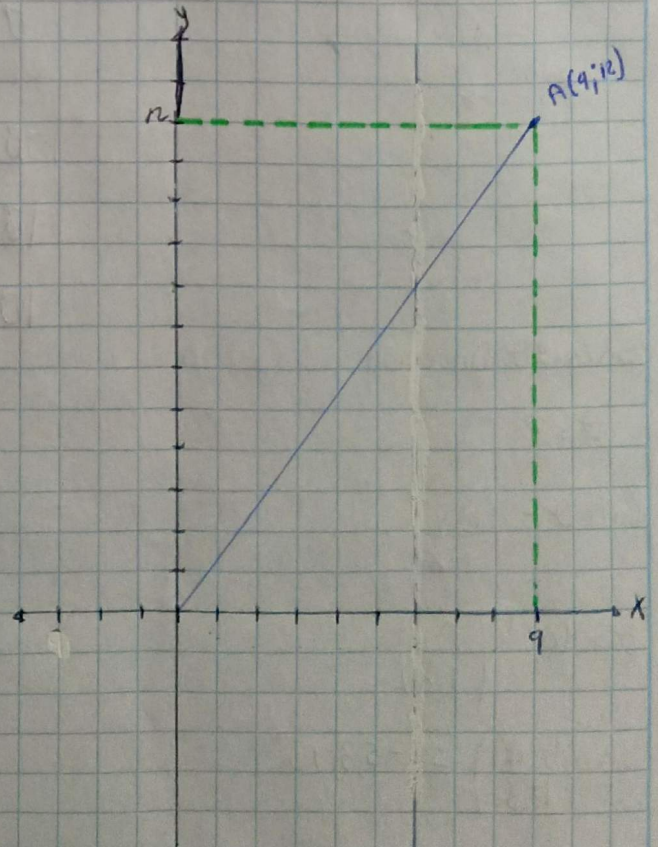
$$\vec{U}_A = \frac{A_x \vec{i} + A_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{9\vec{i}}{15} + \frac{12\vec{j}}{15}$$

$$\vec{U}_A = 0,6\vec{i} + 0,8\vec{j}$$

$$|\vec{U}_A| = \sqrt{(0,6)^2 + (0,8)^2}$$

$$|\vec{U}_A| = 1$$



$\theta = ?$

$$\tan \theta = \frac{A_y}{A_x}$$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

$$\tan^{-1}\left(\frac{12}{9}\right) = 1,33$$

$$\theta = 180 - 1,33 = 178,67^\circ$$

$$\vec{B} = 6\vec{i} + 8\vec{j}$$

$$|\vec{B}| = \sqrt{B_x^2 + B_y^2}$$

$$|\vec{B}| = \sqrt{(6)^2 + (8)^2}$$

$$|\vec{B}| = \sqrt{36 + 64} = \sqrt{100}$$

$$|\vec{B}| = 10 \text{ [m]}$$

Valor Unitario

$$\vec{U}_A = \frac{B_x \vec{i} + B_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{6}{10} + \frac{8}{10}$$

$$\vec{U}_A = 0,6 + 0,8$$

$$|\vec{U}_A| = \sqrt{(0,6)^2 + (0,8)^2}$$

$$|\vec{U}_A| = 1$$

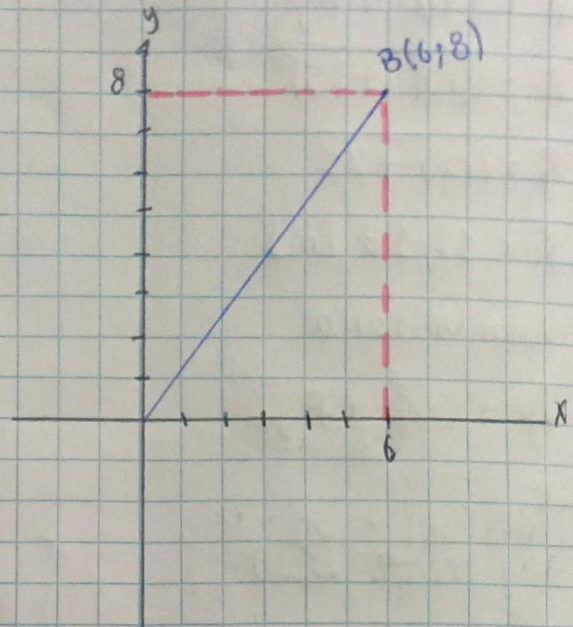
$$\theta = ?$$

$$\tan \theta = \frac{B_y}{B_x}$$

$$\theta = \tan^{-1}\left(\frac{8}{6}\right)$$

$$\theta = 1,33$$

$$\theta = 180 - 1,33 = 178,67^\circ$$



$$\vec{A} = 7\vec{i} + 2\vec{j}$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(7)^2 + (2)^2}$$

$$|\vec{A}| = \sqrt{49 + 4}$$

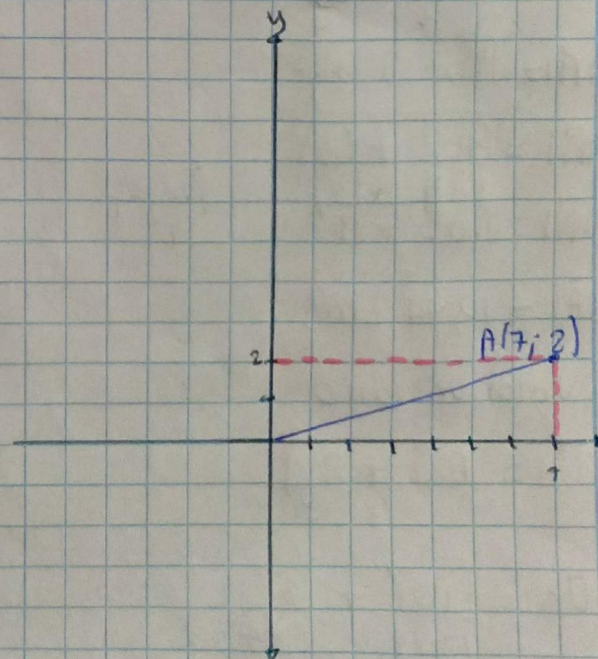
$$|\vec{A}| = \sqrt{53} = 7,28 \text{ [m]}$$

Valor Unitario

$$\vec{U}_A = \frac{A_x \vec{i} + A_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{7}{7,28} + \frac{2}{7,28}$$

$$\vec{U}_A = 0,96 + 0,27$$



$$\vec{A} = 10\vec{i} + 5\vec{j} \text{ [m]}$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(10)^2 + (5)^2}$$

$$|\vec{A}| = \sqrt{100 + 25}$$

$$|\vec{A}| = 11,18 \text{ [m]}$$

Valor Unitario

$$\vec{U}_A = \frac{A_x \vec{i} + A_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{10\vec{i} + 5\vec{j}}{11,18}$$

$$\vec{U}_A = 0,9\vec{i} + 0,4\vec{j}$$

$$|\vec{U}_A| = \sqrt{(0,9)^2 + (0,4)^2}$$

$$|\vec{U}_A| = 1$$

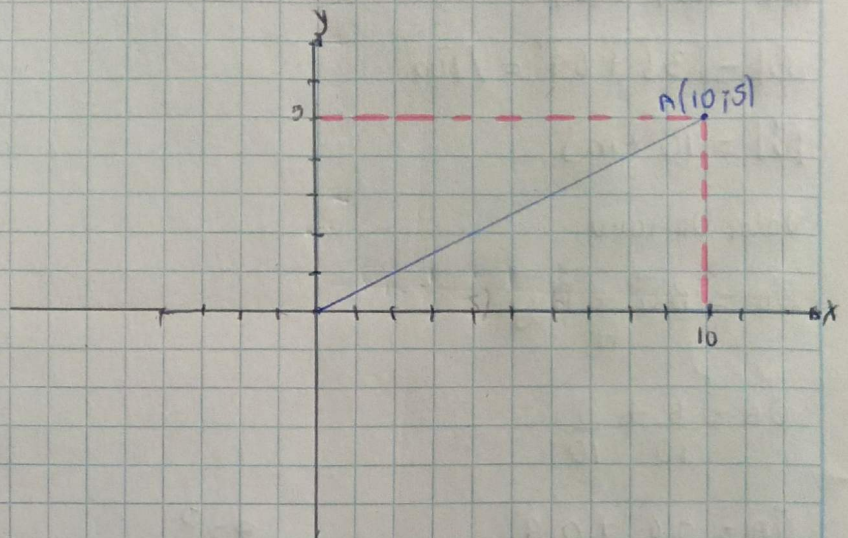
$$\theta = ?$$

$$\tan \theta = \left| \frac{A_y}{A_x} \right|$$

$$\theta = \tan^{-1} \left(\frac{5}{10} \right)$$

$$\theta = 0,5$$

$$\theta = 180 - 179,5^\circ$$



$$\vec{A} = -10\vec{i} + 3\vec{j} \text{ [m]}$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(-10)^2 + (3)^2}$$

$$|\vec{A}| = \sqrt{100 + 9}$$

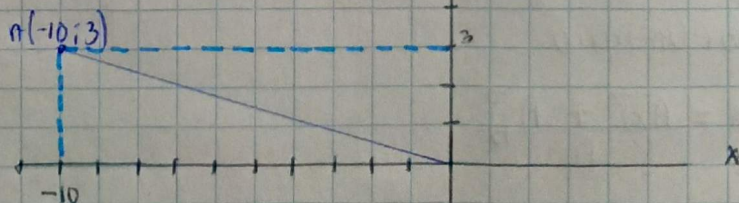
$$|\vec{A}| = 10,44 \text{ [m]}$$

Valor Unitario

$$\vec{U}_A = \frac{A_x \vec{i} + A_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{-10\vec{i} + 3\vec{j}}{10,44}$$

$$\vec{U}_A = -0,96\vec{i} + 0,29\vec{j}$$



$$|\vec{U}_A| = \sqrt{(-0,96)^2 + (0,29)^2}$$

$$|\vec{U}_A| = 1$$

$$\vec{C} = -7\vec{i} - 5\vec{j}$$

$$|\vec{C}| = \sqrt{C_x^2 + C_y^2}$$

$$|\vec{C}| = \sqrt{(-7)^2 + (-5)^2}$$

$$|\vec{C}| = \sqrt{49 + 25}$$

$$|\vec{C}| = 8,6 \text{ [m]}$$

Valor Unitario

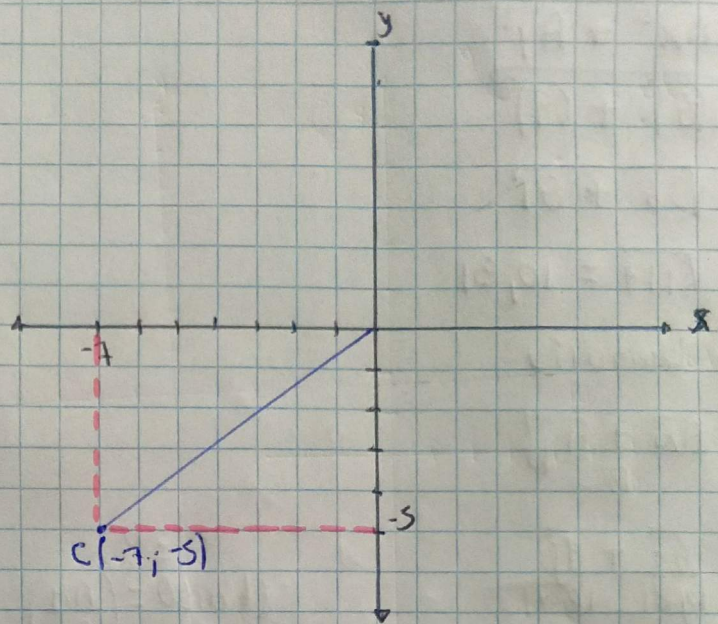
$$\vec{U}_A = \frac{C_x \vec{i} + C_y \vec{j}}{C}$$

$$\vec{C}_A = \frac{(-7)}{8,6} + \frac{(-5)}{8,6}$$

$$\vec{C}_A = -0,81\vec{i} - 0,58\vec{j}$$

$$|\vec{C}_A| = \sqrt{(-0,81)^2 + (-0,58)^2}$$

$$|\vec{C}_A| = 1$$



$$\theta = ?$$

$$\tan \theta = \left(\frac{C_y}{C_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{-5}{-7} \right)$$

$$\theta = 35,53$$

$$\theta = 180^\circ + 35,53 = 215,53^\circ$$

$$\vec{A} = (7\vec{i} + 7\vec{j})$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(7)^2 + (7)^2}$$

$$|\vec{A}| = \sqrt{49 + 49}$$

$$|\vec{A}| = 9,89 \text{ [m]}$$

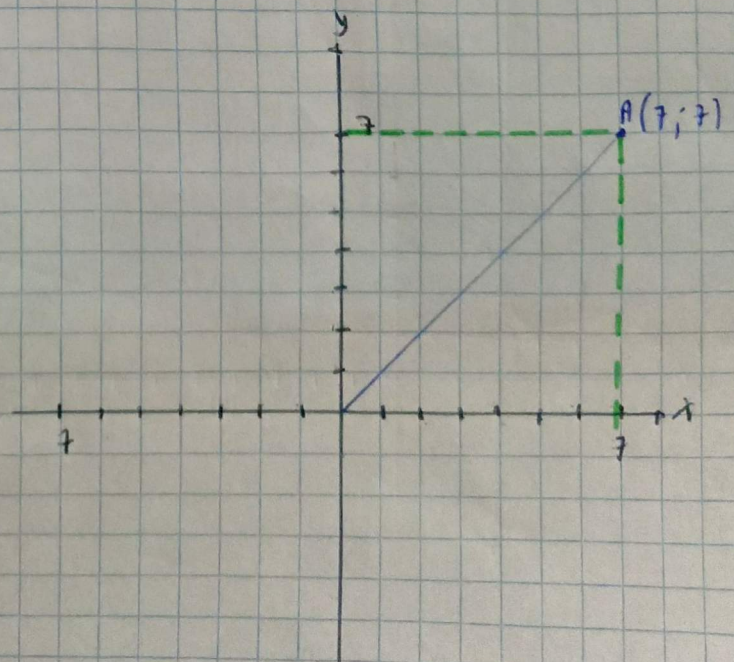
Valor Unitario

$$\vec{U}_A = \frac{A_x \vec{i} + A_y \vec{j}}{A}$$

$$\vec{U}_A = \frac{7}{9,89} + \frac{7}{9,89}$$

$$\vec{U}_A = 0,71 + 0,71$$

$$|\vec{U}_A| = \sqrt{(0,71)^2 + (0,71)^2}$$



$$\vec{B} = 6\vec{i} + 9\vec{j}$$

$$|\vec{A}| = \sqrt{A_x^2 + A_y^2}$$

$$|\vec{A}| = \sqrt{(6)^2 + (9)^2}$$

$$|\vec{A}| = \sqrt{36 + 81}$$

$$|\vec{A}| = \sqrt{117} = 10,81$$

Valor Unitario

$$\vec{U}_A = \frac{A_x\vec{i} + A_y\vec{j}}{A}$$

$$\vec{U}_A = \frac{6\vec{i}}{10,81} + \frac{9\vec{j}}{10,81}$$

$$\vec{U}_A = 0,55 + 0,83$$

$$|\vec{U}_A| = \sqrt{(0,55)^2 + (0,83)^2}$$

$$|\vec{U}_A| = 1$$

$$\theta = ?$$

$$\tan \theta = \left(\frac{A_y}{A_x} \right)$$

$$\theta = \tan^{-1} \left(\frac{9}{6} \right)$$

$$\theta = 56,3$$

$$\theta = 180 - 56,3 = 123,7^\circ$$

